
AI Skills for Kids: Teachable Machine — Train Your Own Image Classifier

Age Group: 10–13 years**Duration: 90 minutes****Difficulty: Intermediate****Setting: Homeschool (1 parent + 1 child)**

1. Session Overview

Today, your child becomes an **AI trainer**! Using a free browser-based tool called *Teachable Machine*, they will teach a computer to recognise different objects using nothing more than a webcam and their own creativity. No coding experience is needed — but by the end of the session, your child will understand core ideas behind how real-world AI systems (like face unlock on phones or spam filters in email) actually learn.

This project is exciting because your child isn't just *using* AI — they are **building** it from scratch, choosing what it learns, watching it make mistakes, and figuring out how to make it smarter. It's hands-on, visual, and endlessly customisable.

2. Learning Objectives

By the end of this session, your child will be able to:

- Explain in their own words what "training a model" means (showing the computer many examples so it can spot patterns).
- Collect and organise labelled image data for at least 3 different categories ("classes").
- Train an image classification model using Google's Teachable Machine tool and test its accuracy live via webcam.
- Identify at least one reason a model makes a wrong prediction (e.g. not enough variety in training images) and try a fix.
- Describe one real-world example of image classification AI they encounter in daily life.

3. Materials & Setup

ITEM	SPEC / NOTES	QUANTITY
Laptop or desktop computer	With a working webcam (built-in or USB), updated Chrome or Firefox browser	1
Internet connection	Stable connection for loading the Teachable Machine web tool	1
Household objects	3–5 small, distinct items for training classes (e.g. a mug, a book, a toy, a fruit, a pair of scissors)	3–5 items
Notebook and pen/pencil	For jotting down class names, predictions, and observations	1 each
Desk lamp or good lighting source	Helps webcam capture clear images	1 (optional but recommended)
Plain wall or blank backdrop	A clutter-free background behind the child while capturing images	1 area
Snack and water	For the break	As needed

Parent setup note: Before the session, open a browser and search for "Teachable Machine" (it's a free tool by Google, no sign-in or download required). Bookmark the page so your child can find it easily. Test that the webcam works and that the browser allows camera permissions.

4. Flexible Schedule (Total: ~90 minutes)

These are suggested ranges — let your child move at their own pace. It's fine to spend more time on the part they enjoy most.

ACTIVITY	SUGGESTED TIME	NOTES
Warm-up chat: What is AI training?	5–10 min	Discussion, no screens yet
Plan your classifier (choose objects/classes)	10–15 min	Notebook planning
Open Teachable Machine & set up project	5 min	Parent check-in on browser/webcam
Collect training images for each class	20–25 min	Most hands-on part
Break	5–10 min	Snack, stretch, move around
Train the model & watch it learn	10–15 min	Parent check-in on results

Test with new objects/poses & refine	15–20 min	Encourage experimentation
Wrap-up reflection & save/share	5–10 min	Discussion + notebook notes

5. Project Breakdown: Teachable Machine — Image Classifier

Step-by-Step Instructions (for your child to read)

1. **Think like a scientist:** Choose 3–5 objects from around the house. Write their names in your notebook — these will be your "classes" (categories the AI will learn to tell apart).
2. **Open the tool:** Go to the Teachable Machine website your parent bookmarked. Click "**Get Started**", then choose "**Image Project**", then "**Standard image model.**"
3. **Name your classes:** You'll see empty boxes labelled "Class 1", "Class 2", etc. Rename them to match your chosen objects (e.g. "Mug", "Book", "Empty").
4. **Parent check-in:** Ask your parent to confirm the webcam is working and you can see yourself in the little camera preview box.
5. **Collect images:** For each class, click "**Webcam**" then **hold down "Hold to Record"** while showing the object from different angles, distances, and lighting. Aim for 50–100 images per class. Don't forget an "Empty" or "Background" class with no object — just the room.
6. **Train the model:** Once all classes have images, click the big "**Train Model**" button. Watch the progress bar — this is the AI studying your images!
7. **Parent check-in:** Once training finishes, sit together and test the live "Preview" panel — hold up each object and see what the AI predicts.
8. **Test and improve:** Try a new angle or a friend holding the object. If the AI gets confused, go back and add more varied images to that class, then retrain.
9. **Reflect:** Write in your notebook: Which class was hardest for the AI to learn? Why do you think that happened?

Exact Tool Steps (Reference "Recipe")

```

STEP 1: Open Teachable Machine → Click "Get Started"
STEP 2: Select "Image Project" → Select "Standard image model"
STEP 3: For each class box:
  - Click the pencil icon → rename class (e.g. "Mug")
  - Click "Webcam" → allow camera access
  - Press and HOLD "Hold to Record" while moving the object
    around: near, far, tilted, different backgrounds
  - Repeat until you have ~50-100 images per class

```

STEP 4: Repeat Step 3 for every class, INCLUDING one
"Background" or "Empty" class (no object, just room)

STEP 5: Click "Train Model" (do not close the tab while training)

STEP 6: When training completes, use the "Preview" panel:

- Hold objects up to the webcam
- Watch the confidence percentage bars update live

STEP 7: If predictions are wrong or shaky:

- Add MORE images to the confused class
- Vary lighting/angle/distance more
- Click "Train Model" again

STEP 8: Optional: Click "Export Model" to save your work

Common Mistakes & How to Fix Them

- **Mistake:** All training images look almost identical (same angle, same spot).
Fix: Move around! Show the object from the side, upside down, close-up, and far away.
- **Mistake:** Forgetting the "Background/Empty" class.
Fix: The AI needs to know what "nothing" looks like too, or it will always guess an object is present.
- **Mistake:** Same messy background behind every object, so the AI learns the background instead of the object.
Fix: Vary the background slightly between recording sessions, or use a plain wall.
- **Mistake:** Testing only with the exact same pose used in training.
Fix: Deliberately test with new angles to see if the AI has really learned, not just memorised.

Success Criteria

Your child has got it when they can:

- Correctly explain why the AI needs many different images per class (not just one photo).
- Get the live webcam preview to correctly identify at least 2 out of 3 objects most of the time.
- Notice on their own when the model is confused and suggest a way to improve it.

6. Conversation Starters & Check-ins

- "What do you think the computer is actually 'looking at' when it decides which object it sees?"
- "Why do you think it helped to add more pictures from different angles?"
- "Can you think of other places in real life where a computer has to tell things apart, like your objects?"

- "What surprised you about how the AI made mistakes?"
- "If you had to teach a friend to do this, what's the one tip you'd give them?"

7. Extensions & Challenges

- **More classes:** Add a 4th or 5th object and see if accuracy drops — why might that happen?
- **Emotion classifier:** Instead of objects, train classes for facial expressions (happy, surprised, neutral).
- **Two-person test:** Have another family member hold the objects to see if the model still works for someone new.
- **Export & explore:** Advanced learners can explore exporting the trained model and see how it could be used in a simple Scratch project (parent can look this up together).
- **Confidence challenge:** Try to get the model's confidence score above 95% for every class by adding more/better training images.

8. Troubleshooting Guide

PROBLEM	WHAT TO TRY
Webcam doesn't show a preview	Check that the browser has permission to use the camera (usually a small pop-up asks this — click "Allow"). Refresh the page if needed.
Training button is greyed out	Make sure every class has at least one image recorded.
Training seems stuck	Training can take 1–3 minutes depending on the computer. Avoid closing the tab or switching windows during this time.
Predictions are inaccurate	Add more images with more variety (angle, lighting, distance) to the class that's being misidentified, then retrain.
Tool won't load at all	Check your internet connection; try refreshing or restarting the browser.
Child is frustrated with "wrong" AI answers	Remind them that mistakes are part of how AI learns — this is exactly what real AI engineers deal with every day!

9. Parent Guidance Tips

- **Let your child click the buttons.** Your role is to sit alongside, not take over the mouse — even if it's slower.
- **Ask, don't tell.** If something isn't working, ask "What do you think is happening?" before offering a solution.

- **Celebrate the "aha" moments** — like when they realise more images = better accuracy. That's the real learning.
- **If frustration builds**, pause for the scheduled break early. A short walk or snack often resets focus quickly.
- **It's okay if the model isn't perfect.** Understanding **why** it's imperfect is more valuable than a flawless result.

10. Follow-up Activities

- Encourage your child to try Teachable Machine's **Audio Project** mode using sounds (like clapping vs. whistling) as a follow-up session.
- Ask your child to spot 3 examples of AI classification in daily life this week (e.g. phone camera recognising faces, email spam filters, voice assistants) and write them in their notebook.
- Challenge them to teach a sibling or friend how to use Teachable Machine, explaining the steps in their own words — teaching others deepens understanding.
- Revisit the project in a few days and try improving the model's accuracy without any help, as an independent challenge.